

NAME: _____

PERIOD: _____

DATE: _____

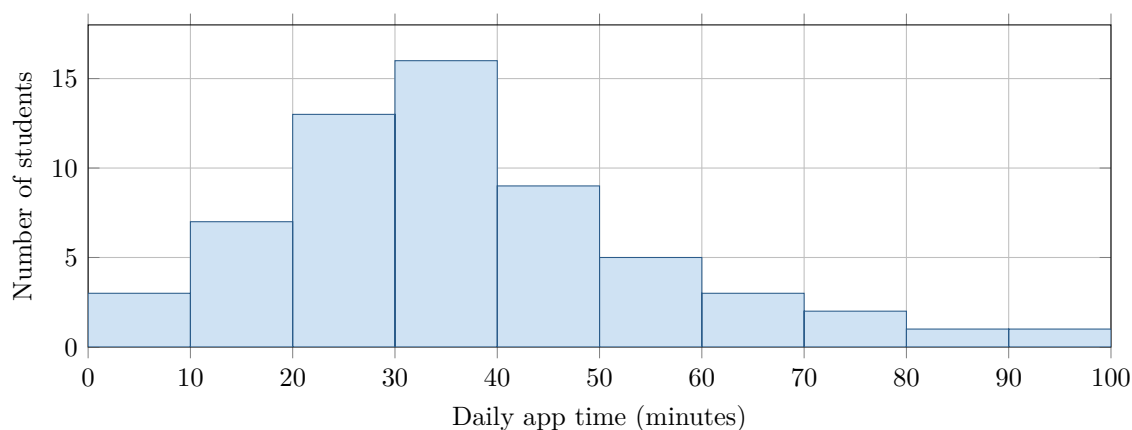
Check the Model

AP Statistics · Unit 2 · Topic 2.11 · B: 2026-11-09 · E: 2026-12-02

[STAR] TODAY'S PROBLEM

A school says daily app time is approximately normal with mean 40 minutes and standard deviation 12 minutes. The histogram shows 60 students; no recorded time was exactly 70 minutes.

Ben: “About 16% should be above 52 minutes.” **Ava:** “For 70 minutes, $z = 2.5$, so less than 1% should be above 70.”



FIRST TAKE — before the video (5 min)

Does the normal model fit? Point to one histogram feature.

[BOOK] USING A NORMAL MODEL (keep on the page)

A normal curve is **mound-shaped and symmetric**. Its parameters are the population mean μ and population standard deviation σ . Check the data's shape before using the model. Empirical rule: about 68% is within 1σ , 95% within 2σ , and 99.7% within 3σ . Standardize with $z = (x - \mu)/\sigma$. A normal-table answer describes the *model*; it describes the data only when the model is appropriate.

Finish after the video:

DOK 1 (a) State μ and σ for the proposed model. What two shape features should approximately normal data have?

DOK 2 (b) Check Ben's and Ava's model calculations. Then find the observed fraction of these 60 students above 70 minutes.

DOK 3 (c) Are the calculations warranted as conclusions about these students? Defend your decision with the histogram's shape and tail count. Explain how the relationship between a model and its data supports your decision, what a reader would conclude about long app times, and what must be checked before using normal methods.

Frame: The calculations are _____ under the model; for these data I would _____ them because _____.

Frame: The model predicts _____ above 70, while the data show _____, so the model _____ the upper tail.

EXIT — turn this in

One thing the video changed about my first take: _____